

1.5 TRIDIAGONAL SYSTEMS OF EQUATIONS

When a large system of linear algebraic equations has a special pattern, such as a tridiagonal pattern, it is usually worthwhile to develop special methods for that unique pattern. There are a number of direct elimination methods for solving systems of linear algebraic equations which have special patterns in the coefficient matrix. These methods are generally very efficient in computer time and storage. Such methods should be considered when the coefficient matrix fits the required pattern, and when computer storage and/or execution time are important. One algorithm that deserves special attention is the algorithm for tridiagonal matrices, often referred to as the Thomas (1949) algorithm. Large tridiagonal systems arise naturally in a number of problems, especially in the numerical solution of differential equations by implicit methods. Consequently, the Thomas algorithm has found a large number of applications.

To derive the Thomas algorithm, let's apply the Gauss elimination procedure to a tridiagonal matrix $\mathbf{T}$, modifying the procedure to eliminate all unnecessary computations involving zeros. Consider the matrix equation:

$$\boxed{\mathbf{T}\mathbf{x} = \mathbf{b}} \quad (1.144)$$

where $\mathbf{T}$ is a tridiagonal matrix. Thus,

$$\mathbf{T} = \begin{bmatrix} a_{11} & a_{12} & 0 & 0 & 0 & \cdots & 0 & 0 & 0 \\ a_{21} & a_{22} & a_{23} & 0 & 0 & \cdots & 0 & 0 & 0 \\ 0 & a_{32} & a_{33} & a_{34} & 0 & \cdots & 0 & 0 & 0 \\ 0 & 0 & a_{43} & a_{44} & a_{45} & \cdots & 0 & 0 & 0 \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ 0 & 0 & 0 & 0 & 0 & \cdots & a_{n-1,n-2} & a_{n-1,n-1} & a_{n-1,n} \\ 0 & 0 & 0 & 0 & 0 & \cdots & 0 & a_{n,n-1} & a_{n,n} \end{bmatrix} \quad (1.145)$$

Since all the elements of column 1 below row 2 are already zero, the only element to be eliminated in row 2 is a_{21} . Thus, replace row 2 by $R_2 - (a_{21}/a_{11})R_1$. Row 2 becomes

$$[0 \quad a_{22} - (a_{21}/a_{11})a_{12} \quad a_{23} \quad 0 \quad 0 \quad \cdots \quad 0 \quad 0 \quad 0] \quad (1.146)$$

Similarly, only a_{32} in column 2 must be eliminated from row 3, only a_{43} in column 3 must be eliminated from row 4, etc. The eliminated element itself does not need to be calculated. In fact, storing the elimination multipliers, $em = (a_{21}/a_{11})$, etc., in place of the eliminated elements allows this procedure to be used as an LU factorization method. Only the diagonal element in each row is affected by the elimination. Elimination in rows 2 to n is accomplished as follows:

$$a_{i,i} = a_{i,i} - (a_{i,i-1}/a_{i-1,i-1})a_{i-1,i} \quad (i = 2, \dots, n) \quad (1.147)$$

Thus, the elimination step involves only $2n$ multiplicative operations to place $\mathbf{T}$ in upper triangular form.

The elements of the $\mathbf{b}$ vector are also affected by the elimination process. The first element b_1 is unchanged, The second element b_2 becomes

$$b_2 = b_2 - (a_{21}/a_{11})b_1 \quad (1.148)$$

Subsequent elements of the $\mathbf{b}$ vector are changed in a similar manner. Processing the $\mathbf{b}$ vector requires only one multiplicative operation, since the elimination multiplier,

$em = (a_{21}/a_{11})$, is already calculated. Thus, the total process of elimination, including the operation on the $\mathbf{b}$ vector, requires only $3n$ multiplicative operations.

The $n \times n$ tridiagonal matrix $\mathbf{T}$ can be stored as an $n \times 3$ matrix $\mathbf{A}'$ since there is no need to store the zeros. The first column of matrix $\mathbf{A}'$, elements $a'_{i,1}$, corresponds to the subdiagonal of matrix $\mathbf{T}$, elements $a_{i,i-1}$. The second column of matrix $\mathbf{A}'$, elements $a'_{i,2}$, corresponds to the diagonal elements of matrix $\mathbf{T}$, elements $a_{i,i}$. The third column of matrix $\mathbf{A}'$, elements $a'_{i,3}$, corresponds to the superdiagonal of matrix $\mathbf{T}$, elements $a_{i,i+1}$. The elements $a'_{1,1}$ and $a'_{n,3}$ do not exist. Thus,

$$\mathbf{A}' = \begin{bmatrix} — & a'_{1,2} & a'_{1,3} \\ a'_{2,1} & a'_{2,2} & a'_{2,3} \\ a'_{3,1} & a'_{3,2} & a'_{3,3} \\ \dots\dots\dots & \dots\dots\dots & \dots\dots\dots \\ a'_{n-1,1} & a'_{n-1,2} & a'_{n-1,3} \\ a'_{n,1} & a'_{n,2} & — \end{bmatrix} \quad (1.149)$$

When the elements of column 1 of matrix $\mathbf{A}'$ are eliminated, that is, the elements $a'_{i,1}$, the elements of column 2 of matrix $\mathbf{A}'$ become

$$a'_{1,2} = a'_{1,2} \quad (1.150a)$$

$$a'_{i,2} = a'_{i,2} - (a'_{i,1}/a'_{i-1,2})a'_{i-1,3} \quad (i = 2, 3, \dots, n) \quad (1.150b)$$

The $\mathbf{b}$ vector is modified as follows:

$$b_1 = b_1 \quad (1.151a)$$

$$b_i = b_i - (a'_{i,1}/a'_{i-1,2})b_{i-1} \quad (i = 2, 3, \dots, n) \quad (1.151b)$$

After $a'_{i,2}$ ($i = 2, 3, \dots, n$) and $\mathbf{b}$ are evaluated, the back substitution step is as follows:

$$x_n = b_n/a'_{n,2} \quad (1.152a)$$

$$x_i = (b_i - a'_{i,3}x_{i+1})/a'_{i,2} \quad (i = n-1, n-2, \dots, 1) \quad (1.152b)$$